2020 RI H2 Mathematics Prelim Paper 2 Solutions

Section A: Pure Mathematics [40 marks]

1	An arithmetic series has first term $\frac{3}{2}$ and fourth term u_4. A geometric series also has first term $\frac{3}{2}$ and fourth term u_4. Given that the common ratio of the geometric series is non-negative and the sum of its first 3 terms is $\frac{21}{2}$, find the sum of the first 10 odd-numbered terms of the arithmetic series. [5]	
1 **[5]**	Given that $u_1 = \frac{3}{2}$, let the common difference of the arithmetic series be d and the common ratio of the geometric series be r. and $\frac{3}{2}+\frac{3}{2}r+\frac{3}{2}r^2 = \frac{21}{2}$ -------- (1) $r^2 + r - 6 = 0$ $(r+3)(r-2) = 0$ $\therefore r = 2,$ since r is non negative. $\frac{3}{2}+3d = \frac{3}{2}r^3$ -----------------(2) Substitute $r = 2$ into (2), $d = \frac{7}{2}$. Sum of first 10 odd numbered terms of AP $S = \frac{10}{2}\left[2\left(\frac{3}{2}\right)+9(7)\right] = 330$	First ten odd-numbered terms refer to the 1st, 3rd, 5th, … , 19th terms not “odd numbers”. The 1st term is $\frac{3}{2}$ and common difference $2d = 7$

2	In a geometric series, the ratio of the sum of the first 8 terms to the sum of the first 4 terms is 17:16. Find the 2 possible values of the common ratio, r_1, r_2 where $r_1 > r_2$. The sum to infinity of the geometric series with first term *a* and common ratio r_1 is denoted by S_1 and the sum to infinity of the geometric series with first term *b* and common ratio r_2 is denoted by S_2. Find $S_1 : S_2$ in terms of *a* and *b*. [5]	
2 **[5]**	$S_4 = \dfrac{a(r^4-1)}{r-1}$ ---------(1) $S_8 = \dfrac{a(r^8-1)}{r-1}$ ---------(2) $\dfrac{(2)}{(1)} \Rightarrow \dfrac{r^8-1}{r^4-1} = \dfrac{17}{16}$ $\dfrac{\left(r^4-1\right)\left(r^4+1\right)}{r^4-1} = \dfrac{17}{16}$ $\therefore r^4 = \dfrac{1}{16} \Rightarrow r_1 = \dfrac{1}{2},\ r_2 = -\dfrac{1}{2}$ When $r_1 = \dfrac{1}{2},\ S_1 = \dfrac{a}{1-\dfrac{1}{2}} = 2a$ When $r_2 = -\dfrac{1}{2},\ S_2 = \dfrac{b}{1+\dfrac{1}{2}} = \dfrac{2}{3}b$ Ratio is $a : \dfrac{1}{3}b$ or $3a : b$	Note sum to n terms for a GP formula.

3	The functions f and g are defined by $$\mathrm{f}(x) = 1 + 3\mathrm{e}^{-x}, \qquad x \in \mathbb{R}, x > 0,$$ $$\mathrm{g}(x) = \lvert x-1 \rvert (x-3), \quad x \in \mathbb{R}, x < c \text{ where } c \text{ is a real constant.}$$	
	(i)	Given that $c = 4$, determine if the composite functions fg and gf exist, justifying your answers. Find the range of the composite function that exists. [4]
	(ii)	Given that g^{-1} exists, state the largest possible value of c. Using this value of c, find $\mathrm{g}^{-1}(x)$. [4]

3(i) **[4]**	$R_g = (-\infty, 3)$ $D_f = (0, \infty)$ Since $R_g \not\subset D_f$, fg does not exist. $R_f = (1, 4)$ $D_g = (-\infty, 4)$ Since $R_f \subseteq D_g$, gf exists. $R_{gf} = [\mathrm{g}(2), \mathrm{g}(4))$ $= [-1, 3)$	State the domain and range for each function clearly. Use proper notation. Note that g attains a minimum at $x = 2$.
(ii) **[4]**	Largest possible value of c is 1. For $x < 1$, $\lvert x-1 \rvert = 1-x$ $\mathrm{g}(x) = (1-x)(x-3)$ $(1-x)(x-3) = y$ $1-(x-2)^2 = y$ $x = 2-\sqrt{1-y} \quad \because x < 1$ $\mathrm{g}^{-1}(x) = 2-\sqrt{1-x},\ x \in \mathbb{R},\ x < 0.$	

4	Given that $y=\mathrm{e}^{\tan^{-1}\left(\frac{x}{2}\right)}$, show that $\left(4+x^2\right)\frac{\mathrm{d}y}{\mathrm{d}x}=2y$. [2]	
	(i)	By repeated differentiation of the above result, find the Maclaurin series for $\mathrm{e}^{\tan^{-1}\left(\frac{x}{2}\right)}$ up to and including the term in x^3. [5]
	(ii)	Hence find the Maclaurin series for $\dfrac{\mathrm{e}^{\tan^{-1}\left(\frac{x}{2}\right)}}{(1+x)^2}$ up to and including the term in x^2. [3]

[2]	$y=\mathrm{e}^{\tan^{-1}\left(\frac{x}{2}\right)}$ $\frac{\mathrm{d}y}{\mathrm{d}x}=\mathrm{e}^{\tan^{-1}\left(\frac{x}{2}\right)}\frac{1}{1+\left(\frac{x}{2}\right)^2}\times\frac{1}{2}$ $=\frac{2\mathrm{e}^{\tan^{-1}\left(\frac{x}{2}\right)}}{4+x^2}$ $\left(4+x^2\right)\frac{\mathrm{d}y}{\mathrm{d}x}=2\mathrm{e}^{\tan^{-1}\left(\frac{x}{2}\right)}$ $\left(4+x^2\right)\frac{\mathrm{d}y}{\mathrm{d}x}=2y$ (Shown)	
(i) **[5]**	$\left(4+x^2\right)\frac{\mathrm{d}y}{\mathrm{d}x}=2y$ ---------- (1) $\left(4+x^2\right)\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+2x\frac{\mathrm{d}y}{\mathrm{d}x}=2\frac{\mathrm{d}y}{\mathrm{d}x}$ ---------- (2) $\left(4+x^2\right)\frac{\mathrm{d}^3y}{\mathrm{d}x^3}+2x\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+2x\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+2\frac{\mathrm{d}y}{\mathrm{d}x}=2\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ $\left(4+x^2\right)\frac{\mathrm{d}^3y}{\mathrm{d}x^3}+4x\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+2\frac{\mathrm{d}y}{\mathrm{d}x}=2\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ ---------- (3) When $x=0$, $y=1$. From equations (1), (2), (3) we get $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{1}{2}$, $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}=\frac{1}{4}$, $\frac{\mathrm{d}^3y}{\mathrm{d}x^3}=-\frac{1}{8}$. By Maclaurin's Theorem,	

	$$e^{\tan^{-1}\left(\frac{x}{2}\right)}=1+\frac{1}{2}x+\frac{1}{4}\left(\frac{x^2}{2!}\right)-\frac{1}{8}\left(\frac{x^3}{3!}\right)$$ $$=1+\frac{1}{2}x+\frac{1}{8}x^2-\frac{1}{48}x^3+\ldots$$	
(ii) **[3]**	$$e^{\tan^{-1}\left(\frac{x}{2}\right)}=1+\frac{1}{2}x+\frac{1}{8}x^2-\frac{1}{48}x^3+\ldots$$ $$\frac{e^{\tan^{-1}\left(\frac{x}{2}\right)}}{(1+x)^2}=\left(1+\frac{1}{2}x+\frac{1}{8}x^2-\ldots\right)(1+x)^{-2}$$ $$=\left(1+\frac{1}{2}x+\frac{1}{8}x^2-\ldots\right)\left(1-2x+3x^2-\ldots\right)$$ $$=1+\left(\frac{1}{2}-2\right)x+\left(\frac{1}{8}-1+3\right)x^2+\ldots$$ $$\frac{e^{\tan^{-1}\left(\frac{x}{2}\right)}}{(1+x)^2}=1-\frac{3}{2}x+\frac{17}{8}x^2+\ldots$$	

5	Referred to the origin O, points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are non-zero and non-parallel vectors. Point C lies on OA, between O and A, such that $OC : CA = 2:1$. Point D lies on OB produced such that $OD : BD = 3:2$.
	(i) Find the position vectors $\overrightarrow{OC}$ and $\overrightarrow{OD}$, giving your answers in terms of $\mathbf{a}$ and $\mathbf{b}$. [2]
	(ii) Show that the point E where the lines BC and AD meet has position vector $\frac{4}{3}\mathbf{a}-\mathbf{b}$. [4]
	(iii) Show that the area of triangle CDE can be written as $k\lvert\mathbf{a}\times\mathbf{b}\rvert$, where k is a constant to be found. [3]
	(iv) It is given that the point F is on BO produced, and OE bisects the angle AOF. Find the ratio $OA : OB$. [3]

(i) [2]	$\overrightarrow{OC}=\frac{2}{3}\mathbf{a}$, $\overrightarrow{OD}=3\mathbf{b}$	
(ii) [4]	BC: $\mathbf{r}=\mathbf{b}+s\left(\frac{2}{3}\mathbf{a}-\mathbf{b}\right)$ AD: $\mathbf{r}=\mathbf{a}+t(3\mathbf{b}-\mathbf{a})$ At E, $\mathbf{b}+s\left(\frac{2}{3}\mathbf{a}-\mathbf{b}\right)=\mathbf{a}+t(3\mathbf{b}-\mathbf{a})$ Since $\mathbf{a}$ and $\mathbf{b}$ are non-zero and non-parallel, $\frac{2}{3}s=1-t \quad \cdots(1)$ $1-s=3t \quad \cdots(2)$ Put (1) into (2): $1-s=3\left(1-\frac{2}{3}s\right)=3-2s$ $s=2$ $\overrightarrow{OE}=\mathbf{b}+2\left(\frac{2}{3}\mathbf{a}-\mathbf{b}\right)=\frac{4}{3}\mathbf{a}-\mathbf{b}$	Use proper notation. Do not write $\ell_{BC}=\mathbf{b}+\lambda...$
(iii)	Area of CDE is	$\mathbf{b}\times\mathbf{a}=-\mathbf{a}\times\mathbf{b}$

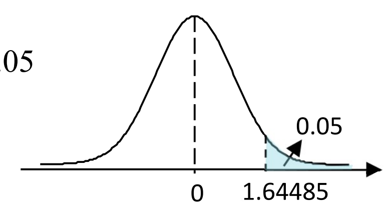
[3]	$\frac{1}{2}\left\|\overrightarrow{DC}\times\overrightarrow{CE}\right\| = \frac{1}{2}\left\|\left(\frac{2}{3}\mathbf{a}-3\mathbf{b}\right)\times\left(\frac{2}{3}\mathbf{a}-\mathbf{b}\right)\right\|$ $= \frac{1}{2}\left\|-\frac{2}{3}\mathbf{a}\times\mathbf{b}+\frac{2}{3}\mathbf{a}\times 3\mathbf{b}\right\|$ since $\mathbf{a}\times\mathbf{a}=\mathbf{b}\times\mathbf{b}=\mathbf{0}$ $= \frac{1}{2}\left\|\frac{4}{3}\mathbf{a}\times\mathbf{b}\right\| = \frac{2}{3}\left\|\mathbf{a}\times\mathbf{b}\right\|$	
(iv) **[3]**	$\overrightarrow{OE} = \frac{4}{3}\mathbf{a}-\mathbf{b}$ is the diagonal of the rhombus with sides $\frac{4}{3}\mathbf{a}$ and $-\mathbf{b}$. So $\left\|\frac{4}{3}\mathbf{a}\right\| = \left\|-\mathbf{b}\right\| \Rightarrow \frac{\left\|\mathbf{a}\right\|}{\left\|\mathbf{b}\right\|} = \frac{3}{4}$, i.e. $OA:OB = 3:4$ Method 2: $\cos\angle AOE = \cos\angle FOE$ $\frac{\mathbf{a}\cdot\left(\frac{4}{3}\mathbf{a}-\mathbf{b}\right)}{\left\|\mathbf{a}\right\|\times OE} = \frac{-\mathbf{b}\cdot\left(\frac{4}{3}\mathbf{a}-\mathbf{b}\right)}{\left\|\mathbf{b}\right\|\times OE}$ $\frac{\frac{4}{3}\left\|\mathbf{a}\right\|^2-\left\|\mathbf{a}\right\|\left\|\mathbf{b}\right\|\cos\theta}{\left\|\mathbf{a}\right\|} = \frac{-\frac{4}{3}\left\|\mathbf{a}\right\|\left\|\mathbf{b}\right\|\cos\theta+\left\|\mathbf{b}\right\|^2}{\left\|\mathbf{b}\right\|}$ where $\theta = \angle AOB$ $\frac{4}{3}\left\|\mathbf{a}\right\|-\left\|\mathbf{b}\right\|\cos\theta = -\frac{4}{3}\left\|\mathbf{a}\right\|\cos\theta+\left\|\mathbf{b}\right\|$ $\frac{4}{3}\left\|\mathbf{a}\right\|(1+\cos\theta) = \left\|\mathbf{b}\right\|(1+\cos\theta)$, $1+\cos\theta\neq 0$ otherwise $\mathbf{a}\parallel\mathbf{b}$ $\frac{\left\|\mathbf{a}\right\|}{\left\|\mathbf{b}\right\|} = \frac{3}{4}$ i.e. $OA:OB = 3:4$ Method 3: $\sin\angle AOE = \sin\angle FOE$ $\frac{\left\|\mathbf{a}\times\left(\frac{4}{3}\mathbf{a}-\mathbf{b}\right)\right\|}{\left\|\mathbf{a}\right\|\times OE} = \frac{\left\|-\mathbf{b}\times\left(\frac{4}{3}\mathbf{a}-\mathbf{b}\right)\right\|}{\left\|\mathbf{b}\right\|\times OE}$ $\frac{\left\|-\mathbf{a}\times\mathbf{b}\right\|}{\left\|\mathbf{a}\right\|} = \frac{\left\|-\frac{4}{3}\mathbf{b}\times\mathbf{a}\right\|}{\left\|\mathbf{b}\right\|}$ $\frac{\left\|\mathbf{a}\right\|}{\left\|\mathbf{b}\right\|} = \frac{\left\|-\mathbf{a}\times\mathbf{b}\right\|}{\frac{4}{3}\left\|-\mathbf{b}\times\mathbf{a}\right\|} = \frac{3}{4}$, so $OA:OB = 3:4$	**Note alternative solution methods.**

Section B: Probability and Statistics [60 marks]

6	For events X and Y, it is given that $\mathrm{P}(X \cap Y') = \frac{1}{2}$, $\mathrm{P}(X \cup Y') = \frac{3}{4}$ and $\mathrm{P}(X \mid Y') = \frac{50}{63}$. Find	
	(i)	$\mathrm{P}(Y')$, [2]
	(ii)	$\mathrm{P}(X)$, [2]
	(iii)	$\mathrm{P}(X \cap Y)$ and state with a reason whether X and Y are independent events. [3]

(i) **[2]**	$\mathrm{P}(X \mid Y') = \dfrac{\mathrm{P}(X \cap Y')}{\mathrm{P}(Y')}$ $\mathrm{P}(Y') = \dfrac{\mathrm{P}(X \cap Y')}{\mathrm{P}(X \mid Y')}$ $= \dfrac{\left(1/2\right)}{\left(50/63\right)}$ $= \dfrac{63}{100}$	
(ii) **[2]**	$\mathrm{P}(X \cup Y') = \mathrm{P}(X) + \mathrm{P}(Y') - \mathrm{P}(X \cap Y')$ $\mathrm{P}(X) = \mathrm{P}(X \cup Y') + \mathrm{P}(X \cap Y') - \mathrm{P}(Y')$ $= \dfrac{3}{4} + \dfrac{1}{2} - \dfrac{63}{100}$ $\mathrm{P}(X) = \dfrac{31}{50}$	
(iii) **[3]**	$\mathrm{P}(X) = \mathrm{P}(X \cap Y) + \mathrm{P}(X \cap Y')$ $\mathrm{P}(X \cap Y) = \mathrm{P}(X) - \mathrm{P}(X \cap Y')$ $= \dfrac{31}{50} - \dfrac{1}{2}$ $= \dfrac{3}{25}$ Since $\mathrm{P}(X \cap Y) = \dfrac{3}{25} \neq \dfrac{31}{50} \times \dfrac{37}{100} = \mathrm{P}(X) \times \mathrm{P}(Y)$, X and Y are not independent events. OR Since $\mathrm{P}(X) = \dfrac{31}{50} \neq \dfrac{50}{63} = \mathrm{P}(X \mid Y')$, X and Y' are not independent events So X and Y are not independent events.	

7	In a factory, machines pack sugar into bags of 1 kg each on average, with variance σ^2 kg^2. The manufacturer is concerned that the machines are putting too much sugar into the bags and decides to carry out a hypothesis test. A random sample of 8 bags are selected and their total mass is 8.4 kg.	
	(i)	Stating a necessary assumption, carry out a test of the manufacturer's concern at the 5% significance level if $\sigma = 0.08$. [5]
	(ii)	Use an algebraic method to calculate the range of values of σ^2 for which the null hypothesis would not be rejected at the 5% significance level. [3]

7(i) **[5]**	Let X be the mass of a bag of sugar in kg. The necessary assumption is X follows a normal distribution. $\bar{x} = \frac{8.4}{8} = 1.05$ $\text{H}_0: \ \mu = 1$ vs $\text{H}_1: \ \mu > 1$ Perform a 1-tail test at 5% significance level. Under H$_0$, $\bar{X} \sim \text{N}\left(1, \frac{0.08^2}{8}\right)$ Using a z-test, p-value $= \text{P}\left(\bar{X} > 1.05\right) = 0.0385$ (3 s.f) Since p-value $= 0.0385 \leq 0.05$, we **reject** H_0 and conclude that there is **sufficient** evidence, at the 5% significance level, to support the manufacturer's concern.	Use proper notation. Present solution argument clearly.
(ii) **[3]**	Under H$_0$, $\bar{X} \sim \text{N}\left(1, \frac{\sigma^2}{8}\right)$ $p\text{-value} > 0.05$ $\text{P}\left(\bar{X} \geq 1.05\right) > 0.05$ $\text{P}(\left(Z \geq \frac{1.05-1}{\sqrt{\sigma^2/8}}\right) > 0.05$ From GC, $\text{P}((Z \geq 1.64485) = 0.05$ From the diagram,  $0.05\sqrt{\frac{8}{\sigma^2}} < 1.64485$ $\sigma^2 > 8\left(\frac{0.05}{1.64485}\right)^2$ $\sigma^2 > 0.00739$ (3 s.f)	

8	In this question you should state clearly the values of the parameters of any normal distribution you use. In a supermarket, the masses in grams of apples have the distribution $\mathrm{N}(90, 13^2)$ and the masses in grams of potatoes have the distribution $\mathrm{N}(170, 30^2)$.	
	(i)	Find the probability that the mass of a randomly chosen potato is more than twice the mass of a randomly chosen apple. [3]
	A certain salad recipe requires 5 apples and 6 potatoes.	
	(ii)	Find the probability that the total mass of 5 randomly chosen apples and 6 randomly chosen potatoes is between 1.2 and 1.5 kilograms. [3]
	The salad recipe requires the apples and potatoes to be prepared by peeling and slicing them. The process reduces the mass of each apple by 15% and the mass of each potato by 25%.	
	(iii)	Find the probability that the total mass, after preparation, of 5 randomly chosen apples and 6 randomly chosen potatoes is not more than 1.2 kilograms. [3]

(i) **[3]**	Let X and Y be the masses in grams of a randomly chosen apple and a randomly chosen potato in grams respectively. i.e. $X \sim \mathrm{N}(90, 13^2)$, $Y \sim \mathrm{N}(170, 30^2)$. $\mathrm{E}(Y-2X) = \mathrm{E}(Y) - 2\mathrm{E}(X) = 170 - 2(90) = -10$ $\mathrm{Var}(Y-2X) = \mathrm{Var}(Y) + 2^2\,\mathrm{Var}(X) = 30^2 + 2^2(13^2) = 1576$ $Y - 2X \sim \mathrm{N}(-10, 1576)$ Required probability $= \mathrm{P}(Y > 2X)$ $= \mathrm{P}(Y - 2X > 0)$ $= 0.401$ (3 s.f.)	**Define random variables clearly.**
(ii) **[3]**	Let $T = X_1 + X_2 + \ldots + X_5 + Y_1 + Y_2 + \ldots + Y_6$. $\mathrm{E}(T) = 5\mathrm{E}(X) + 6\mathrm{E}(Y) = 5(90) + 6(170) = 1470$ $\mathrm{Var}(T) = 5\mathrm{Var}(X) + 6\mathrm{Var}(Y) = 5(13^2) + 6(30^2) = 6245$ $T \sim \mathrm{N}(1470, 6245)$ Required probability $= \mathrm{P}(1200 < T < 1500)$ $= 0.648$ (3 s.f.)	Note that $X_1 + \ldots + X_5 + Y_1 + \ldots + Y_6$ $\neq 5X + 6Y$ State the distribution of T

(iii) **[3]**	Let $W = 0.85\left(X_1 + X_2 + ... + X_5\right) + 0.75\left(Y_1 + Y_2 + ... + Y_6\right)$ $\mathrm{E}\left(W\right) = 0.85 \times 5\mathrm{E}\left(X\right) + 0.75 \times 6\mathrm{E}\left(Y\right)$ $= (0.85)(5)(90) + (0.75)(6)(170) = 1147.5$ $\mathrm{Var}\left(W\right) = 0.85^2 \times 5\mathrm{Var}\left(X\right) + 0.75^2 \times 6\mathrm{Var}\left(Y\right)$ $= (0.85^2)(5)(13^2) + (0.75^2)(6)(30^2) = 3648.0125$ $W \sim \mathrm{N}\left(1147.5,\ 3648.0125\right)$ Required probability $= \mathrm{P}(W \le 1200)$ $= 0.808$ (3 s.f.)	

9	The continuous random variable X has the distribution $N(\mu,\sigma^2)$. It is known that $P(X<k)=0.2$ and $P(X<7)=0.8$.	
	(i)	Show that $P(k<X<7)=0.6$ and write down the value of $P(\mu<X<7)$. [2]
	(ii)	Express μ in terms of k. [1]
	You may use $\sigma^2=12$ for the rest of the question.	
	(iii)	Show that $\mu=4.0845$ correct to 4 decimal places. [3]
	(iv)	Find $P(\lvert X\rvert<k)$. [2]
	It is given that $2P(X\le r)=3P(X>r)$ for a certain constant, r.	
	(v)	Ten independent observations of X are randomly selected. Find the probability that there are more observations of X with values greater than r than observations of X with values less than r in the selection. [4]

(i) **[2]**	$P(k<X<7)=P(X<7)-P(X<k)=0.8-0.2=0.6$ (shown) $P(\mu<X<7)=0.3$	
(ii) **[1]**	Since $P(k<X<\mu)=P(\mu<X<7)=0.3$, by symmetry $\mu=\dfrac{k+7}{2}$.	**Draw a diagram!**
(iii) **[2]**	$P(X<7)=0.8\Rightarrow P(Z<\dfrac{7-\mu}{\sqrt{12}})=0.8$ Therefore $\dfrac{7-\mu}{\sqrt{12}}=0.8416$ (4 d.p) $\dfrac{7-\mu}{\sqrt{12}}=0.8416\Rightarrow\mu=4.0845$	
(iv) **[3]**	$k=2\mu-7=1.169$ $P(\lvert X\rvert<k)=0.135$ (3 s.f)	
(v) **[4]**	$2P(X\le r)=3P(X>r)$ $2\left(1-P(X>r)\right)=3P(X>r)$ $5P(X>r)=2$ $\Rightarrow P(X>r)=0.4$ Let Y be the number of observations out of 10 with values greater than r. $Y\sim B(10,0.4)$ $P(Y\ge 6)=1-P(Y\le 5)=0.166$ (3s.f)	

10	A group of 13 people consists of 6 single men and 5 single women and a married couple. A committee of 7 is to be selected from the group.	
	(i)	Find the number of committees that can be formed if there is no restriction in the selection. [1]
	(ii)	Show that there are 658 such committees with more women than men. [2]
	Given that a committee of 7 people is to be selected from the group such that the committee contains more women than men,	
	(iii)	find the probability of getting a committee that consists of 4 single women and 3 single men. [2]
	(iv)	find the probability of getting a committee that contains at least one married member. [2]
	The 4 women and 3 men who were finally selected for the committee included the married couple. The 7 members sit at random around a table with 7 chairs.	
	(v)	Find the probability that the men are all separated from each other. [2]
	(vi)	Given that the men are all separated from each other, find the probability that the married couple sit next to each other. [3]

(i) **[1]**	Number of committees if there is no restrictions in the selection $= {}^{13}C_7 = 1716$	
(i) **[2]**	Case 1: 4 women 3 men Number of committees = ${}^{6}C_4 \times {}^{7}C_3 = 525$ Case 2: 5 women 2 men Number of committees = ${}^{6}C_5 \times {}^{7}C_2 = 126$ Case 3: 6 women 1 man Number of committees = ${}^{6}C_6 \times {}^{7}C_1 = 7$ Total number of committees = 658 (Shown)	
(iii) **[1]**	P(the committee will consist of 4 single women and 3 single men) $= \frac{{}^{5}C_4 \times {}^{6}C_3}{658} = \frac{100}{658} = \frac{50}{329}$	
(iv) **[2]**	Number of committees with no married member $= {}^{5}C_4 \times {}^{6}C_3 + {}^{5}C_5 \times {}^{6}C_2$ $= 100 + 15$ $= 115$	

	Number of committees with at least one married member $= 658 - 115$ $= 543$ P(the committee will contain at least one married member) $= \frac{543}{658}$	
	Alternative Solution We consider 3 cases: Case 1: Husband in, wife out Number of committees $= {}^5C_4 \times {}^6C_2 + {}^5C_5 \times {}^6C_1 = 75 + 6 = 81$ Case 2: Wife in, husband out Number of committees $= {}^5C_3 \times {}^6C_3 + {}^5C_4 \times {}^6C_2 + {}^5C_5 \times {}^6C_1 = 200 + 75 + 6 = 281$ Case 3: Both husband and wife are in Number of committees $= {}^5C_3 \times {}^6C_2 + {}^5C_4 \times {}^6C_1 + {}^5C_5 = 150 + 30 + 1 = 181$ Total number of such committees = 543 P(the committee will contain at least one married member) $= \frac{543}{658}$	
(v) **[2]**	Number of ways to sit 4 women around the table = 3! Number of ways to slot the 3 men = $4 \times 3 \times 2 = 24$ Number of sitting arrangements where the all the men are separated $= 3! \times 24$ $= 144$ Required probability $= \frac{144}{6!}$ $= \frac{1}{5}$	
(vi) **[3]**	Number of ways to sit 4 women around the table = 3! Number of ways to sit the man next to his wife = 2 Number of ways to sit the other 2 men = ${}^3P_2$ Number of ways to have all the men separated from each other and the married couple sit next to each other $= 3! \times 2 \times 3 \times 2 = 72$ Required probability $= \frac{72}{144}$ $= \frac{1}{2}$	

11 In conjunction with the Great Singapore Sale, a certain electronics store is having a lucky draw for their customers. In each round of the lucky draw, the customer draws two balls randomly, one after another, with replacement from a box containing 20 red balls, 30 blue balls and 50 white balls. The colour of each ball drawn is noted and points are awarded accordingly as follows.

Colour of ball	Point(s)
Red	5
Blue	4
White	1

The customer's score in each round of the lucky draw is the total number of points awarded for the balls drawn. To illustrate, a customer scores a total of 5 points if he draws a blue ball and a white ball in a round of the lucky draw, regardless of the order of appearance of the balls.
You may assume that the 100 balls are indistinguishable from each other apart from their colour.

Let X denote the total number of points scored by a customer in a round of the lucky draw.

(i) Tabulate the probability distribution of X. [3]

(ii) Find $\mathrm{E}(X)$ and show that $\mathrm{Var}(X) = 6.02$. [2]

Mr Lim participated in 50 rounds of the lucky draw.

(iii) Using a suitable approximation, find the probability that Mr Lim's average score is at least 6. [3]

A customer wins a cash voucher if his total score for one round of the lucky draw is more than 6.

The total number of cash vouchers won by Mr Tan in n rounds of the lucky draw is denoted by Y.

(iv) Find the least value of n such that there is a probability of more than 0.7 that Mr Tan will win more than 3 cash vouchers in total. [4]

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[2]</td>
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<table>
<tr><td>x</td><td>2</td><td>5</td><td>6</td><td>8</td><td>9</td><td>10</td></tr>
<tr><td>$\mathrm{P}(X=x)$</td><td>$\frac{1}{4}$
=0.25</td><td>$\frac{3}{10}$
=0.3</td><td>$\frac{1}{5}$
=0.2</td><td>$\frac{9}{100}$
=0.09</td><td>$\frac{3}{25}$
=0.12</td><td>$\frac{1}{25}$
=0.04</td></tr>
</table>
</td>
<td>“tabulate”</td>
</tr>
<tr>
<td>(ii)
[3]</td>
<td>Using GC, $\mathrm{E}(X) = 5.4$

$\mathrm{E}\left(X^2\right) = 35.18$

$\mathrm{Var}\left(X\right) = \mathrm{E}\left(X^2\right) - \left[\mathrm{E}\left(X\right)\right]^2 = 35.18 - (5.4)^2 = 6.02$ (shown)</td>
<td>GC is to be used here.</td>
</tr>
<tr>
<td>(iii)
[3]</td>
<td>Since $n = 50$ is large, by Central Limit Theorem,

$\overline{X} \sim \mathrm{N}\left(5.4, \frac{6.02}{50}\right)$ approximately.

Required probability $= \mathrm{P}\left(\overline{X} \geq 6\right) = 0.0419$ (3 s.f.)</td>
<td>State distribution of $\overline{X}$</td>
</tr>
<tr>
<td>(iv)
[4]</td>
<td>$\mathrm{P}\left(\text{winning a cash voucher}\right)$
$= \mathrm{P}\left(X > 6\right)$
$= \mathrm{P}\left(X = 8, 9 \text{ or } 10\right)$
$= \frac{9}{100} + \frac{3}{25} + \frac{1}{25} = \frac{1}{4}$

$\therefore Y \sim \mathrm{B}\left(n, \frac{1}{4}\right)$
We must have
$\mathrm{P}\left(Y > 3\right) > 0.7$
$\Rightarrow 1 - \mathrm{P}\left(Y \leq 3\right) > 0.7$
$\Rightarrow \mathrm{P}\left(Y \leq 3\right) < 0.3$
Using GC,
<table>
<tr><td>n</td><td>$\mathrm{P}\left(Y \leq 3\right)$</td></tr>
<tr><td>18</td><td>0.30569</td></tr>
<tr><td>19</td><td>0.26309</td></tr>
</table>
$\therefore$ Least $n = 19$</td>
<td>Show sufficient working to justify answer.</td>
</tr>
</table>